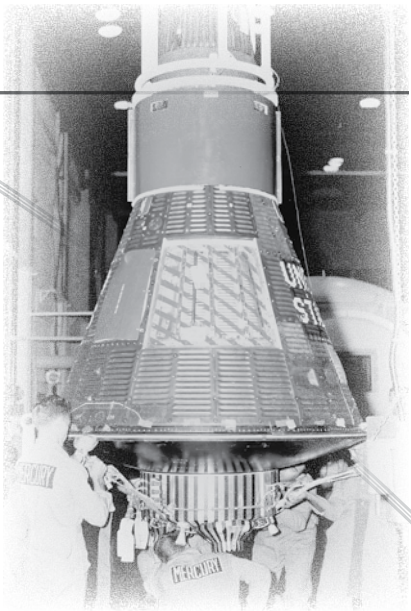
CHAPTER 3



THE MERCURY SPACE FLIGHT NETWORK

By the summer of 1961, the GSFC had fully established itself as the lead NASA center for directing science application satellite projects along with the communication, tracking, and data network that support them. In a little over two years since its founding as the first new field center established after the creation of NASA, GSFC's workforce had grown from 216 to 1,900, accounting for 11 percent of the space agency's total.[1] The STADAN was quite busy. It quickly matured during this time, supporting a wide array of satellite projects ranging from the original Vanguard and Explorer series to the newer TV and weather satellites such as TIROS (Television Infrared Observation Satellite) and even privately built communication satellites. The first satellite project which Goddard assumed full responsibility for was Explorer 6, launched on 7 August 1959. This began 16 years of GSFC association on this very successful series, which continued until 1981, with the orbital decay of Explorer 55. In April 1960, TIROS 1 downlinked the first ever global cloud-cover photographs taken from orbit. Solar science satellites were launched beginning with Solrad 1 in June 1960. AT&T's Telstar 1, the world's first commercial satellite, was launched on 10 July 1962.

The first Orbiting Solar Observatory (OSO 1) was launched on 7 March 1962. For years, John C. Lindsay, a former Associate Chief of the Space Sciences Division at Goddard, had investigated satellite designs using spin stabilization which enabled observations of the Sun in ways not possible from the surface of Earth. By 1960, the Ball Brothers Aerospace Corporation of Boulder, Colorado (at the time the leading developer and provider of stabilized pointing control devices for sounding rockets and balloon systems) had produced a successful engineering prototype of a spin-stabilized satellite. OSO1 produced many new findings. Among them was the discovery that the Sun's corona has openings (now called coronal holes) which were interpreted as huge, fast-moving bubbles rising through the corona.[2]

Even as application and science satellites were hitting full stride and making ever more exciting discoveries on a regular basis, events, though, were taking place which soon redefined NASA's priorities and led to the establishment of an all-together, different Goddard spaceflight network.

★ ★ ★

On 12 April 1961, the Soviet Union launched Yuri A. Gagarin, a 27-year-old Second Lieutenant in the Soviet Air Force, into orbit on Vostok (East) 1.[3] But unlike Sputnik, Gagarin's flight did not come as a total shock to the United States. What the Soviet Union accomplished four years earlier with Sputnik was the first sign to the international community of the existence of a full-fledged and very robust space program, one in which they proceeded with at an unrelenting pace. As early as the mid-1950s, a disturbing picture of Russia's growing capability in long-range, heavy payload rocketry emerged. The CIA, using the Lockheed U-2 high-altitude reconnaissance aircraft, had reported in 1957 the first Soviet ICBM on its launch pad at the Baikonur-Tyuratam space complex. Over the next two years, the world would know what the CIA had long suspected: the USSR was embarking on a series of space experiments designed to put a human into space.

In May 1960, the Soviet news agency Tass reported the launching of a "Spacecraft 1" into orbit. It weighed a massive 10,008-pounds and contained a "dummy cosmonaut." This was followed by more qualification flights carrying dogs which took place over the next several months. By the following April, the veil of secrecy surrounding Soviet space activities was rapidly breaking down.

It began with a rumor from Moscow on 7 April that a cosmonaut had been launched in secret. This was followed by a message wired to London on midnight of 11 April quoting unidentified sources as saying that a cosmonaut had returned from space and was undergoing physical examination for illness, and he was "suffering from post-flight effects of a nature more emotional than physical." Variations of this story immediately surfaced,

including one by a French correspondent who suggested that the mystery cosmonaut was none other than Vladimir Ilyushin, son of the famous Soviet aircraft designer. The White House immediately denied the authenticity of these reports saying that the United States had no evidence whatsoever that such a flight had taken place. However, the next day, Moscow released the following official announcement:

> The world's first spaceship, Vostok, with a man on board was launched into orbit from the Soviet Union on 12 April 1961. The pilot space-navigator of the satellite-spaceship Vostok is a citizen of the USSR, Flight Major Yuri Gagarin. The launching of the multi-stage space rocket was successful and, after attaining the first escape velocity and the separation of the last stage of the carrier rocket, the spaceship went into free flight on a round-the-Earth orbit. According to the preliminary data, the period of revolution of the satellite-spacecraft round the Earth is 89.1 minutes. The minimum distance from the Earth at perigee is 175 km [109 miles] and the maximum at apogee is 302 [188 miles], and the angle of inclination of the orbit plane to the Equator is 65° 4'. The spacecraft with the navigator weighs 4,725 kg [10,420 pounds] excluding the weight of the final stage of the carrier rocket.[4]

Although the actual implementation of a U.S. man-in-space program had to await the creation of NASA, concept proposals for "manned-satellites" and how to track them had been in circulation as far back as the early 1950s. These took on a more serious form in 1956 when the Air Force embarked on a study known as "Project 7969" entitled, *A Manned Ballistic Rocket Research System*, designed specifically to investigate the requirements for a human orbiting laboratory. The formulation of ideas in that project did not directly translate into an actual program, but it did push the National Advisory Committee for Aeronautics to begin its own studies of human spaceflight.

The Secretary of Defense created, in early 1958, an Advanced Research Projects Agency (ARPA) to coordinate civilian and military research for piloted and unpiloted space projects, the idea being that ARPA would act as an interim space agency until Congress passed the appropriate legislation establishing NASA. In this capacity, ARPA articulated the first objectives of the American human space program. The goal was clear: "Achieve at the earliest practicable date orbital flight and successful recovery of a manned satellite, and to investigate the capabilities of man in this environment."[5] These goals were to be embodied in Project Mercury, America's first human spaceflight program.

Within the newly created space agency, the Langley Research Center located in Hampton, Virginia near the mouth of the Chesapeake Bay, was the home to two key groups entrusted with starting Project Mercury. The

STG was one of these, responsible for the overall implementation of the man-in-space program, formally designated as Project Mercury on 26 November 1958. The STG directed the development of both the launch vehicle and the spacecraft as well as the recruiting of flight crews and flight controllers. Another cadre of personnel formed the other group, informally called the Tracking System Study Group (TSSG). The TSSG helped the STG establish network support requirements for Project Mercury. Heading it was Edmond C. Buckley, who was later to be in charge of all NASA tracking activities as the Agency's first Associate Administrator for Tracking and Data Acquisition (from 1962 to 1968).

The STG was at the time under the jurisdiction of the newly established Space Projects Center. In 1961, this center moved to Greenbelt to form the nucleus of the new GSFC. By fall of that year, however, the STG had moved on to Houston, Texas to establish the MSC. With this move, MSC immediately became the lead NASA center for all human spaceflight activities and assumed overall responsibility for executing Project Mercury. The job of implementing an effective worldwide tracking and data acquisition network remained with GSFC. These initial shifts in center responsibilities, as it turned out, determined how each of these Field Centers eventually ended up supporting NASA's human spaceflight activities for the next 40 years, roles that continue today.[6]

One of the principals on the TSSG was H. William Wood. A member of Ed Buckley's team at Langley, Bill Wood was investigating X-ray and gamma ray sensing instruments for possible use aboard spacecraft when he got the call to help plan a network for Project Mercury. Said Wood:

> At Langley, we worked under the supervision of Hartley Soule, an assistant laboratory director who helped us cut through the red tape and made us realize early on that we were engaged in a very unique and challenging undertaking. However, the Langley Center did not want to become distracted from its primary role as a research organization. . . . In my opinion, it did not wish to evolve into an operational hub for space activities. Thus tracking, data acquisition and computer activities for Project Mercury were assigned to the Goddard Space Flight Center prior to the first manned orbital flight.[7]

Wood and others who agreed to help were told by Soule that they could return to Langley whenever they wished. Those who agreed to consider a transfer soon met at the new Space Projects Center with Director Harry J. Goett. Some ended up staying at Langley, and others transferred with the STG to the MSC in Houston.



America's first human spacecraft was Mercury. It flew six astronauts into space (two sub-orbital, four orbital) from May 1961 to May 1963. Weighing some 1,360 kilograms (3,000 pounds), the vehicle was launched into space atop the U.S. Army Redstone rocket and later, the Air Force Atlas-D intercontinental ballistic missile. Pictured is Wally Schirra's Mercury 8 capsule *Sigma 7* being readied in Hanger S at Cape Canaveral in September 1962. (NASA Image Number GPN-2000-001441)

The TSSG began addressing the network issue in earnest in the fall of 1958. The job soon turned out to be bigger than anyone had anticipated. Network requirements for human spaceflight were rather different than those needed to support unmanned satellites. Unlike the Minitrack Network—which passively received spacecraft signals and did not make an accurate orbit determination until several orbits had been complete—human spacecraft tracking had a much more *instantaneous contact requirement*. Towards that end, radar tracking rather than interferometry, was essential. Telemetry now had to be augmented, for example, to monitor the health of the astronaut. The complexity of command functions also greatly increased over what was needed for science satellites in order to ensure the safe orbiting and, more importantly, return of the crew. Finally, a manned spaceflight network was obviously needed to establish two-way voice communication between the ground and the vehicle.[8]

By the end of the year, the TSSG had identified three major areas of concern associated with a Mercury tracking network, concerns which had up until then been widely underestimated. First, and somewhat to their surprise, the group found that there was no such thing in existence as either a commercial or a military, real-time, worldwide communication system. Here was a vital ingredient of mission control that was completely missing! Second, there was no existence of reliable, high-capacity data links that could carry the large amounts of data between the computer facilities and the mission control center. Third, good radars were available, but they were designed primarily to track DOD ballistic missiles and reentry vehicles, their beams too narrow to expeditiously locate and keep track of a fast-moving spacecraft in orbit.[9]

The Space Task Group began to feel that the weight of the network job was diverting too much attention from its primary job of designing the Mercury spacecraft. In January 1959, Charles W. Mathews, a member of the STG, recommended to Abe Silverstein, Director of Space Flight Development at NASA Headquarters, that the STG be formally relieved of the responsibility for building the network and that the responsibility be given to the TSSG. This was a seemingly reasonable request since the TSSG had already been studying the network problem for nearly a year. On 16 February, Silverstein formally directed this change in a memorandum to J. W. Crowley, Director of Aeronautical and Space Research at NASA Headquarters.

Silverstein's memorandum asked the TSSG to "complete and refine" network plans so as to satisfy requirements generated by the STG and to "place and supervise" contracts for generating procurement specifications and a final network deployment. Langley officials were to make use, wherever practical, of DOD personnel and facilities in the Pacific Missile Range (PMR) in Point Mugu, California; the Atlantic Missile Range (AMR) in Cape Canaveral, Florida; the White Sands Missile Range (WSMR) in Alamogordo, New Mexico; and the Eglin Gulf Test Range (EGTR), Eglin

Air Force Base, Florida. In this manner, the practice of NASA leveraging DOD assets wherever practical began early on in the construction of the Mercury Space Flight Network (MSFN).

On 16 February 1959, the Langley network study group was officially renamed the Tracking And Ground Instrumentation Unit, or TAGIU. Two years later, TAGIU would be relocated to GSFC when the new Center opened. Leadership of TAGIU was assigned to Barry Graves who had been part of the TSSG since the beginning. Several key participants from TSSG also continued their assignments on the new team. Leading their respective disciplines were:

George B. Graves, Jr., *Electronics and Head of TAGIU*
James J. Donegan, *Computers*
Ray W. Hooker, *Site Selection and A&E (Architecture and Engineering)*
Paul Vavra, *Assistant to Graves*
H. William Wood, *Logistics*

TAGIU quickly grew to 35 people. This nucleus was in essence heir to Langley's radar and high-speed flight experience, plus six months' worth of network studies. Although TAGIU engineers knew roughly what a Mercury Network should entail, they did not yet have the detailed specifications needed to procure hardware and to begin building up stations. Getting industry help under contract and enlisting the aid of the Air Force was among the first items of business. (The USAF was at the time building up their own network: the Air Force Satellite Control Network.) TAGIU's effort went from turning studies into requirements, requirements into specifications, specifications into hardware, and finally, hardware into stations. Even though the major functions of the Mercury Network had already been decided on: tracking, telemetry reception, voice communication, and capsule command; the team fully realized that these functions could be performed in a number of different ways.

By early spring 1959, TAGIU was ready to generate detailed network specifications. To this end, four contracts to industry were awarded:

1 Ford Aeronutronics, *Study radar coverage and trajectory computation requirements*

2 MIT Lincoln Laboratories, *General consultation and proposal evaluation*

3 RCA Service Corporation, *Write the network specifications*

4 Space Electronics, *Design the Mission Control Center*

Things moved quickly. Said Wood:

> We were told that we had about two years to complete the job; then we would return to basic research. Of course, that is not the way it turned out. . . . I was told one morning that I would be managing the RCA contract and that same afternoon we had our first meeting with the contractor. We moved pretty fast! At the time, a great deal of attention was being given to the Mercury spacecraft itself: how to get a man into orbit and how to return him safely to Earth. But very little attention had been given to communications and tracking which had to support those flights. So we were asked to get busy and develop a set of specifications. This we did by 1959. They were no thicker than an issue of the *Aviation Week* magazine.[10]

The network developed around 12 ground rules:

1 The launch azimuth of approximately 32.5° was fixed early on. It enabled maximum use of DOD range facilities and kept the ground track of the spacecraft over the continental United States much of the time to preclude over-flight of countries that might not cooperate. It also resulted in acceptable recovery areas close to U.S. Naval facilities.

2 The Atlantic Missile Range based at Cape Canaveral was to be employed for launch and recovery operations. This was an easy decision because of AMR's already superb facilities and experience.

3 The network was to be worldwide, using stations in foreign countries where necessary, and operate on a real-time basis to monitor spacecraft status and astronaut health.

4 The space medicine community strongly advocated continuous voice contact with the astronaut, but this requirement proved impractical. Despite the controversy, STG and TAGIU moved ahead using the goal of a maximum 10 minute, loss-of-signal and voice contact in between stations (the so-called 10-minute "dead time" rule).[11]

5 A centralized control center was to be built, but controllers would be located at each ground station in case of communications difficulties.

6 Proven tracking and communications equipment was to be employed to the extent possible with a minimum of research and nonrecurring engineering development.

7 In order to enable a go/no-go decision to be made, and to effect emergency reentry prior to orbit insertion before the spacecraft approached the African land mass, continuous tracking from network radars downrange of the launch site was needed to give the computers enough real-time data for accurate orbit prediction before the spacecraft passed beyond the Bermuda tracking station.

8 After orbit injection, intermittent tracking was sufficient. Continuous tracking was again to be required during deorbit and reentry so as to accurately pinpoint the splashdown point.

9 These existing radars would need acquisition aids because of their narrow beamwidths.

10 Only computers could cope with the flood of tracking and telemetry data arriving from all the network stations. A centralized computer facility with data link to the Mission Control Center would therefore be needed.

11 Frequent network simulations and exercises would be needed to train the operators.

12 Redundancy would be required to provide the reliability needed for manned flight.

The team started with the assumption that the network would support a three-orbit Mercury mission at an inclination of about 33° to the Equator. Next, it determined just how long a communication gap could be tolerated between stations. The first design iteration had 20 stations scattered around the globe; eventually, the number was reduced to 18. The procedure was definitely quite straight forward. As Wood put it, the team literally just "took [out] a world atlas and tentatively located these sites."[12]

By early 1960, site surveys for the entire network had been conducted and the building of the MSFN began. During Mercury, Mission Control was located at Cape Canaveral, Florida, adjacent to the launch site, while the network computer center was located at Goddard. All the computer

activities were handled on a remote basis with Goddard computers, for example, to drive the displays and plot-boards at the Cape. (This would be the case until 1965, when starting with Gemini 4, the Manned Spacecraft Center in Houston acquired the necessary computer capability for human flight operations when the new Mission Control Center opened.)

Satellite programs and STADAN had already given GSFC quite a head start, providing the expertise and experience needed for network communications, simulations and data processing. Furthermore, Goddard management under Center Director Harry Goett desperately wanted the work. GSFC could claim, with justification, that to expand the staff and facilities at Greenbelt rather than build new ones at Langley or Houston was more economical and technically feasible. Some thought was given by Headquarters to locating the Mercury ground computers at Cape Canaveral. But the Cape was already heavily involved in Air Force and Navy launches. In addition, NASA wanted to keep its computing facility separate from the DOD. These factors all made Goddard the most logical choice to lead this work.

On a Mercury flight, the network computer facility would receive reams of data from its field stations. Jack Mengel, who four years earlier led the development of the Minitrack Network and who, along with most of his original Naval Research Laboratory staff now at Goddard, was asked by Graves to develop the MSFN computing capability. This made sense since Mengel's group was still in the throes of expanding their computing facilities and had just taken over the Vanguard Computation Center formerly under ownership of the Naval Research Laboratory. Before long, some 60 Goddard and IBM contractors were assigned to support the MSFN under Niles R. "Buck" Heller who became Chief of the Manned Space Flight Support Division at Goddard.[13]

The network relied on proven C-band and S-band radars for tracking; for redundancy, some stations used both. Radar skin-tracking (which uses direct reflections of radio waves transmitted from the ground to the target) while fine for tracking ballistic missiles, was not considered reliable enough at orbital altitudes to be used in the Mercury program. Instead, two radar-triggered onboard beacons (or transponders) at these wavelengths were located on the spacecraft to provide the ground with more robust radar returns.

The C-band FPS-16 and the S-band VERLORT (Very Long Range Tracking) radar built by the Reeves Instrument Corporation were specifically stipulated by the TAGIU, as both were considered proven systems, having been used for years by the DOD. Manufactured by RCA and used by the Army at White Sands since 1957, the FPS-16 was a 3.6-meter (12-foot) dish, high-precision radar capable of tracking to an accuracy of seven meters at a range of 925 kilometers (500 nautical miles). The VERLORT, a 3-meter (10-foot) transportable dish operated by trailer-housed equipment, had a slightly longer range of 1,300 kilometers (700 nautical miles). It was a radar developed in World War II for tracking aircraft. Limited in range and

accuracy, it did have a reputation for being able to acquire an object with great reliability and begin tracking in a very short amount of time.

The only new piece of major tracking equipment specified was an Active Acquisition Aid (AAA) required for spacecraft acquisition by the narrow-beam radars. The acquisition antenna was actually an array of 18 small antennas mounted on a screen. By locking onto a radio signal transmitted from the spacecraft, this arrangement allowed it to begin coarse tracking to an accuracy of about half a degree. This pointing information was then made immediately available via a cable and switching system called the "Acquisition Bus" to aim the other antennas. The AAA, made by the Cubic Corporation, had a very wide beam of 20°, effective out to 1,500 kilometers (800 nautical miles).[14]

Proven equipment was also stipulated for other elements of the ground-to-space link, which included telemetry, command, and voice. TAGIU selected systems that were already available or, as a minimum, not too difficult to adapt from existing hardware. Telemetry from the Mercury spacecraft was downlinked at 226.2 MHz using the analog modulation technique of Pulse Amplitude Modulation (PAM)/Frequency Modulation (FM), a scheme commonly used for ballistic missile and aircraft testing work. Two separate telemetry links between network sites and the spacecraft were utilized for redundancy. Each station had dual-redundant FRW-2 transmitters for command uplink. These were commonly used by the military for high performance aircraft testing. The FRWs were connected to a 10 KW high power amplifier to give them the needed range to send commands to a spacecraft in low-Earth orbit. This capability enabled ground controllers at the MSFN sites to issue commands to a spacecraft in the event the astronauts were to somehow become incapacitated. Common VHF radio was used for two-way, air-to-ground voice communication; a backup was operated on the international military aircraft emergency frequency.[15]

A seminal ground rule that TAGIU stipulated was that nonrecurring engineering costs for equipment research and development could be minimized through reliance on off-the-shelf hardware. Radars, radios and computers existed, but no one had ever tried to mold them together into a cohesive, real-time network on a global scale. Not unlike a jigsaw puzzle, when TAGIU tried to put the electronic puzzle together, it found that most of the available pieces did not fit well together. And most did not have the requisite reliability to support human spaceflight. Modification of available equipment also became a frustrating problem. Modifications were often made after equipment had been fielded just to bring them up to MSFN standards. Equipment that could not be modified had to be backed up with 100 percent spares to ensure operating redundancy. Network equipment reassigned from the DOD was considerably more reliable than common off-the-shelf equipment, not surprising as military specifications were more rigorous and based on operating environments similar

to those of the MSFN. However, obtaining priority military equipment was, for the most part, difficult due to availability.[16]

Before the MSFN could be used to support a mission, it of course had to be checked out, first on a station by station basis and then as an integrated network. One of the most valuable testing methods during network shakedown was simulated spacecraft tracking using aircraft flyovers. Although ground stations had a collimation tower at the edge of the antenna field, it was used only to get a first order calibration. Calibration airplanes, or "cal planes," flying at 7,600 meters (25,000 feet) were needed for final system calibration and verification of an antenna's ability to actually track a moving object. Graves had asked Bill Wood to develop a "real world" method to accept the stations after they were constructed. Wood's people came up with a variety of ideas, ranging from using balloons to devices mounted on towers. Each had its technical advantage as well as disadvantage, and there was really no clear-cut choice. During initial discussions at Goddard, there were, in fact, as many votes against using calibration planes as there were for it. When Barry Graves finally pressed for a recommendation, Wood would later say, "I suppose in self defense, I said airplanes."[17] To do this, NASA procured a Douglas DC-3, two DC-4s and later a Lockheed L-1649A Super Constellation (known both affectionately, and sometimes not so affectionately by those who had to track them, as "Connie") and outfitted them with Mercury TT&C flight-qualified electronics.[18]

Calibration aircraft were used in different ways to test the status of network stations. For STADAN, they were used to calibrate the Minitrack system and to conduct acceptance testing for new antennas. The DC-3 and DC-4 primarily supported the STADAN while the L-1649A supported the MSFN. Calibration of the Minitrack sites for instance, was performed by flying at 7,600 meters and making East-West crossings of the main Minitrack beam in 10 percent steps north and south of the station zenith. The RF (at 108MHz, and later, 136MHz) crossings of the main beam were electronically compared with the positions obtained from the flashing lamp on the belly of the aircraft and measured against the stellar background. Flyovers were done for a period of up to six weeks prior to a scheduled mission and usually took a couple of hours. The aircraft made multiple passes at predetermined altitudes, speeds and directions to simulate a spacecraft passing overhead. The antenna would attempt to autotrack the calplane during each pass. These station simulations ("sims") were designed to uncover any hidden anomalies and to help prepare station operators for their upcoming mission. Unlike a spacecraft orbiting Earth, however, the calibration airplanes did not fly in well-defined trajectories. As such, they actually turned out to be more difficult to track than the spacecraft. After a few exercises, the prevailing thought was that "if you could track the cal plane, you could track the real thing."[19]

Since the DC-3 was not pressurized, the crew had to go on oxygen bottles when flying over 3,600 meters (12,000 feet). Later, Hal Hoff at

Goddard required all NASA members of the calibration team to undergo high altitude chamber orientation at Andrews Air Force Base. This turned out to be smart. Former Goddard engineer Dave Harris recalled an episode where a crew member monitoring the transmission equipment suffered from a classic case of oxygen deprivation and the response of the others "was quite rapid."[20] There were also simulated ditching sessions held to prepare for any contingency. Besides serving as a flying target for tracking tests, another benefit of these calibration planes was to ferry necessities to the foreign sites. Toilet paper was a frequent cargo when heading to South America.

In addition to these site-specific exercises, computer-based simulations were used to exercise the entire network. The principal tool of computerized simulation was the Computation and Data Flow Integrated Subsystem, or CADFISS, developed by James Donegan and Goddard's IBM contractors. During an "integrated-sim," CADFISS interrogated the various stations via the Mercury ground communications network (and later the NASCOM) while GSFC computers analyzed the response and determined the status of each element in the dataflow. In this way, stations could be checked simultaneously, faults isolated and debriefings held with Goddard engineers directly "over the loop" immediately after a simulation session. As a mission date approached, these integrated-simulations were conducted with increasing frequency.[21]

★ ★ ★

When the STG was officially formed on 7 October 1958, it fell under the jurisdiction of GSFC rather than Langley because NASA Headquarters had every intention of moving the group to the new Greenbelt Center. As the human spaceflight program expanded, however, it became apparent that Goddard would no longer be able to absorb STG responsibilities while still carrying out its prime responsibility of directing unmanned satellite programs. It was also evident that human spaceflight was going to continue well beyond just Project Mercury and that it required a new center of its own. Due to its very charter and the work it was already doing on Mercury, the STG was identified as the nucleus of the proposed MSC and in the fall of 1961, the group literally picked up and moved to Houston (the location having recently been selected by Capitol Hill as the site for the new Field Center).

The official word on delegating MSFN network responsibility to GSFC was given in a 3 April 1961 letter from Abe Silverstein, Head of the Office of Space Flight Programs at Headquarters, to Harry J. Goett, the first Director of Goddard, entitled "Mercury Network Operating Responsibilities." On 12 May 1961, Langley center director Floyd L. Thompson and Harry Goett met at Greenbelt to arrange the transfer of network operating responsibility. They established a committee (with a Langley chairman) to oversee the transfer

which was scheduled to take place during the third quarter of calendar year 1961. To help meet this new directive, Goett brought in Ozro M. Covington to Goddard from the WSMR where he had been the Technical Director of the Army Signal Missile Support Agency. His experience in organizing the White Sands range instrumentation was exactly what Goett was looking for, experience that soon proved invaluable in the buildup of the MSFN.[22] This was a strategic move on Goett's part as Covington brought with him a very-abled manager named Henry Thompson from White Sands to serve as his deputy.[23] Thompson, in turn, recruited George Q. Clark from White Sands. (Clark later became the first Tracking and Data Relay Satellite System Project Manager at GSFC, responsible for both the pioneering technical solution and the innovative procurement approach. See Chapter 7.)

In addition to his tangible technical abilities, Covington had unique intangibles which were greatly needed to soothe rifts that immediately began developing between Greenbelt and Houston. It was in the midst of these organizational changes that "turf wars" between the two NASA Field Centers appeared regarding ownership and the role that the network was going to play in the upcoming human space program. Commenting on this touchy issue, Covington recalled:

> During those early days there were many debates and conflicts as to the assignment of the manned spaceflight program. There was a time when the Goddard Center was to be its base of operation. The center had already been charged with major unmanned flight programs, all of which also required sophisticated tracking and communications support. But soon it became evident that a separate NASA organization had to be created to plan and direct America's man-in-space efforts. It was, after all, a mammoth undertaking, particularly after President John F. Kennedy had told the world that the United States would send a man to the Moon and return him safely to Earth in the 1960–1970 decade. Indeed, this was not the only conflict as to who would be responsible for the manned missions. There were similar conflicts as to who would be charged with the difficult and challenging tracking and communications tasks supporting these flights. There was the U.S. Air Force which was launching and tracking its missiles from Cape Canaveral. It certainly wanted the job, and always there was the question of the wisdom of separating the mission control center and the network. As Goddard's representative for manned space flight, I attended many of those early planning sessions, and it was during one of those meetings that I had occasion to spend some two hours with Dr. Christopher Kraft who was then planning some of the early Mercury missions. We met in a hotel room

> near Cape Canaveral and discussed at length the difficulties arising from the separation of mission and network responsibilities. We developed a very productive relationship which continued throughout the years, ranging from the Mercury and Gemini through the Apollo programs.[24]

The "Not Invented Here" syndrome was by no means unique to GSFC and MSC. In developing the Mercury Network and its Control Center, it was clear early on that NASA was conducting a very different kind of launch than what the Cape was used to. Besides the obvious presence of having a man on board, the payloads that NASA was launching were generally active in all phases of a launch, from prelaunch countdown through insertion into orbit. This was generally not the case, though, for DOD payloads which were mostly dormant and were powered up only after orbit insertion. NASA thus had to install its own network equipment at the Cape and at key downrange stations. The DOD and its contractors were naturally not very happy when TAGIU insisted that all equipment used to interface directly with the manned payload were to be designed and controlled by MSFN engineers and that they be operated in accordance with NASA procedures. Happy or not, there was really no other way to reliably implement the network.[25]

There was also not always complete agreement as to the GSFC role when it came to manned tracking. Goett questioned the wisdom of committing his people and resources to a major program not entirely under his control. But Covington would explain years later, Goett came to see it the other way:

> When he saw the impressive team we had assembled for the tracking job, he agreed and relented. 'Let's go after it', he agreed, and the Center did indeed, becom[ing] the nerve center of NASA's worldwide manned spaceflight tracking network, just as it was also supporting major unmanned missions in the near-Earth regions. We assembled a team of some 350 top flight people, supported by a large number of contractor employees. Goddard's administrative staff under Dr. Michael J. Vaccaro gave us the critical support services we required to tackle the job, from personnel recruiting to budget planning and contract administration. We needed this support very badly.[26]

Covington, for one, realized the importance of open communications between the two Centers and treated MSC as a customer rather than a rival:

> I considered it to be my primary function to maintain the closest possible liaison with our 'customer', the Manned [Spacecraft] Center in Houston. We created a reliable system of tracking sta-

> tions around the globe along with communications links and computer facilities all designed to provide the mission controllers with the information they needed to get our men into space.[27]

At the conclusion of Project Gemini in 1967, Covington was named Goddard's Director for Manned Flight Support, overseeing all network activities in preparation for the Apollo Moon program. The management structure at NASA Headquarters also shifted during this time so as to meet the changing requirements of human spaceflight. Since leaving Langley in early 1958, Edmond C. Buckley, as Director of Space Flight Operations, had headed NASA's network operations, reporting directly to Abe Silverstein. To streamline operations, one of the first things Buckley did was to consolidate all of NASA's tracking and data networks (STADAN, MSFN and DSN) under him to form the Office of Tracking and Data Acquisition (OTDA) on 1 November 1961.[28]

On 21 May 1959, Graves and his team sent out requests for proposals formally soliciting industry bids to build a manned spaceflight tracking network for NASA. On 30 June, a team led by Western Electric Company, Inc. was awarded the prime contract. Of the roughly $80 million spent in constructing the MSFN, about 85 percent ($68 million) went to the Western Electric team. The company named Rod Goetchius as Program Manager. In addition to Western Electric, three major subcontractors were on the team. Their roles were:

> Western Electric: Overall program management, procurement, production, transportation, installation and testing of equipment. Design and implementation of the ground communication subsystem. Training of maintenance and operating personnel.
>
> Bell Telephone System: Analysis and development of operation plans and tests. Design of command and control displays at Cape Canaveral and Bermuda. (Bell selected Stromberg-Carlson to build and install the flight control displays.) Provide a simulation system for flight controllers and astronauts.
>
> Bendix Corporation: Design and fabricate telemetry and tracking display equipment. Systems design, fabrication and integration of radars not already furnished by the government. (Bendix obtained new radars from RCA and Reeves Instrument Corporation.) Design and fabrication of all Mercury spacecraft communication equipment.
>
> IBM: Computer programming and operations at GSFC and Bermuda. Maintenance and operation of the launch and display subsystem at Cape Canaveral.[29]

Construction of the Mercury Space Flight Network ground stations proceeded swiftly in 1960 and 1961. Site surveys had established a set of minimum requirements for geographical locations:

> No physical obstructions to the transmission and reception of signals greater than 1° elevation angle
>
> Existence of adequate separation distance between receiving and transmitting antenna to prevent electromagnetic and physical interference
>
> Minimum outside radio interference (a quiet RF location)
>
> Existence of housing and utilities
>
> Availability of good roads[30]

One of the first stations to be established on foreign soil was also one of the most critical. With the exception of Cape Canaveral, Bermuda, some 1,450 kilometers (900 miles) due east of the Carolina coast, was the most complex and important of the 15 MSFN ground stations. This was because the flight path of the Mercury Atlas took it almost directly over the island, providing a short but crucial 25 second window to track and make decisions on its status during ascent into orbit.

As Bill Wood explained, the station was important at the time for these and several other reasons:

> The first station to be definitely decided upon was Bermuda. It was needed because of the high failure rate of the Atlas booster which in those early days was about 50 percent, and since an abort situation was a highly probable reality, we needed Bermuda to keep an eye on the Cape Canaveral launches and the first critical phases of the flight downrange. We planned a control center there as well as at the Cape to give us reliable communications and controls should we be forced to make abort decisions. We were, after all, dealing with manned missions and unproven, even unreliable communications.[31]

Lynwood C. "Lyn" Dunseith, one of the original MSFN engineers who in 1982 became the Director of the Data Systems and Analysis Directorate at the Johnson Space Center, also recalled that in those days:

> We had great difficulties with our communications from Bermuda, a key station during the launch phase of any mission. So, a control center was established there in the event Bermuda had to enter the picture for launch and abort decisions.[32]

The schedule was tight. On 15 March 1961, less than two months before Alan Shepard's scheduled suborbital flight, Washington finally reached a formal agreement with the United Kingdom of Great Britain and Northern Ireland to operate a tracking station on the island.[33] The main site was on Cooper's Island, a small 77-acre rock-coral shelf just off of Saint David's Island on the northern shores of Bermuda. The station was located on an eastward extension of Kindley Air Force Base, on land already managed by the United States Air Force. Its use dated back to a World War II agreement between President Roosevelt and Prime Minister Churchill. Another site was in Town Hill on the main island. The station cost $5 million to build in 1961.[34]

NASA employed a large workforce to operate this station—60 contractors along with 20 Bermudians. Since the tracking network (as well as the space agency itself) was still young, everyone was put to work immediately. Robert E. Spearing, who today is the Director of Space Communications at NASA Headquarters, was 22-years-old when he began working at the GSFC to help get Bermuda up and running. He recalled:

> When I started, I was placed with another engineer who was fairly experienced working on what we call UHF command systems. . . . These are high power transmitters that are used to actually send information to the spacecraft and are also used for what we call command destruct range safety options. About three or four months after I started, the fellow I was working with informed me that he was leaving the Agency for another job. I, of course, went to see his supervisor and asked him what I should do since this fellow was leaving. He looked up at me and he said 'You're it!' . . . That is the way NASA operated back in those days. We were not heavily populated. We were definitely not overstaffed and everyone was given a lot of responsibility to get things done in the best way you could.[35]

The criticality of the station to make the key go/no-go call was not lost on the Bermudians. When construction began in April 1961, the local island newspaper headlined "Bermuda Tracking Station Has Vital Part to Play in First Manned U.S. Spaceflight," going on to boast that "an order from here could bring it down."[36] For fiscal reasons (along with perhaps more importantly, diplomatic expediency and promotion of international goodwill), maximum use of the local workforce was exercised in Bermuda. This worked out well, for

the most part, not only in Bermuda but in other places as well. Some problems did arise, particularly in nonindustrialized countries where work was mostly done by hand. Even in more advanced places such as Australia, contractors often ran into "specifications and standards" discrepancies where, for instance, electromechanical devices made in the USA did not interface correctly with local, European-based standards like voltages and the imperial versus metric systems—something that any traveler can relate to even today.[37]

Even as construction was underway on Bermuda, an agreement was reached to build another critical station, this one in the South Pacific. On 6 April 1961, Secretary of State for Foreign Affairs, H. C. Hainworth, exchanged signatures with Walworth Barbour, Charge d'Affaires ad interim



Designated as Site No. 11, the Canton Island tracking station was located four and one-half miles southeast of the airport terminal in an area used as a fighter strip during World War II. Designed only for telemetry and voice communications, the site was rather sparse with no radar or spacecraft command capability. Communication was maintained with the Mission Control Center by the Goddard teletype and voice loop networks while communication with the Hawaiian area was maintained over a teletype circuit and by a limited traffic single sideband circuit. The island was staffed by 47 U.S. contractors. (Unnumbered photograph. Folder # 8810, NASA Historical Reference Collection, NASA History Division, NASA Headquarters, Washington DC.)

of the British government, formally establishing a tracking station on the small island of Canton. This was the culmination of planning by NASA which first started in 1959. When LRC was first delegated by Headquarters with the responsibility for establishing the Mercury network, a chief aim was to identify locations that maximized communication coverage over three orbits, this being the then planned duration of the Mercury orbital missions.

As the TAGIU examined the ground track map, it was found that a large gap existed from the time when the spacecraft left the Australian station at Woomera to when signals could be acquired by the Hawaii Station on Kauai. During the first orbit, the ground track of the Mercury capsule took it well south of the Hawaiian Islands such that acquisition-of-signal would not have been possible until the capsule was over the North American continent—a communication gap of over 25 minutes! A loss-of-signal of this duration was not considered at all acceptable when Mercury was first being planned (even though longer quiet periods were to become common place on later missions). A site had to be found to close this gap. It turned out that the ground track of the Mercury spacecraft took it almost directly over Canton Island in the Kiribati Republic.

Canton (also known by its Kiribati name of Kanton or Abariringa) is the largest and most northern of the Phoenix Islands, located in the middle of the Pacific Ocean just south of the Equator. It is a volcanic atoll, made of a low, narrow rim of land surrounding a large shallow lagoon. As with most atolls in this region, it is relatively small, only seven kilometers (four and one-half miles) wide on the west, from which it narrows to a southeast point some 14.5 kilometers (nine miles) away. Since its discovery by independent sailors in the early 1800s, mostly American whalers, the island served mainly as a stopping point for American and British ships traversing the Pacific shipping lanes. Canton broke into the news in 1937 when American and New Zealand astronomers chose it as a spot from which to view the total solar eclipse of 8 July; enough publicity was generated to at least put the tiny spot on the map. Prior to this, about the only news coming out of the island was the continual British efforts to reassert their jurisdiction over the Phoenix Islands and the speculated role Canton may have played in the disappearance of Amelia Earhart and her navigator Fred Noonan. In 1938 and 1939, Pan American Airways developed an extensive airport on the island, deepening and clearing the lagoon to initiate air travel to New Zealand using Canton as one of their ports of call.

On 9 August 1959, NASA Administrator Keith Glennan wrote a letter to the Department of the Interior proposing to establish a tracking station on the island. (At the time, the United States had codominion status along with the United Kingdom for the island.) In a reply to Mr. Glennan, Interior Secretary Fredrick A. Seaton granted permission, saying "We wish to assure you of any further cooperation your Administration may require of this Department in furtherance of this most important project."[38] In December, Langley sent a del-

egation to the island for a site survey and to begin negotiations with the local authorities. Construction began the next year, and the station was well prepared by the time John Glenn orbited Earth on 20 February 1962.

Located 3,200 kilometers (2,000 miles) northeast of Canton are the Hawaiian Islands. Situated in the middle of the Pacific Ocean, Hawaii provided an ideal setting for a network ground station, picking up spacecraft as they emerged from the South Pacific. Construction of the Kokee Park Tracking Station on the southwest hills of Kauai began in May 1960. Eleven months later, the station was completed, coming online in time to support the first unmanned orbital Mercury Atlas test flight (MA-4) in September 1961. Five months later, Glenn completed the first U.S. human orbital mission on MA-6.[39]

Even though Hawaii was designed to support the human spaceflight program, it was also frequently tasked to support Goddard's unmanned science satellites because of its good location and full compliment of equipment. In fact, multitasking of this station went beyond NASA. Hawaii was a shared effort between the space agency and the Navy's Pacific Missile Range (PMR). Beginning with its construction, the Kokee Station was a joint venture between the two departments, with the Navy operating a tracking and data acquisition facility on the grounds of what had been Bonham Air Force Base on the western side of the island. When construction was finished in August of 1961, it was integrated into the PMR command at Kaneohe, Oahu. In this arrangement, the Barking Sands (U.S. Navy) and Kokee (NASA) facilities operated as a single, integrated station even though they were separated by some 3.5 kilometers (2.2 miles) and 1,200 meters (4,000 feet) in elevation.[40]

In the spring of 1965, the DOD transferred control of the facility to the Air Force Western Test Range, headquartered at Vandenberg Air Force Base in Lompoc, California. According to Virg True, former Station Director, there was little change as result of this reorganization other than adding one more scheduling office into the mix. True himself, though, transferred to the Air Force in order to remain in what he felt was the more exciting space program as opposed to the Navy's missile programs. Such was the appeal of human spaceflight in the 1960s.[41]

It should come as little surprise that squabbles existed between NASA and the Air Force. Disagreements often existed on things ranging from operations control and information flow to daily responsibilities, particularly during human spaceflight operations. At Hawaii, these disagreements quickly magnified in a short amount of time soon after the station became operational. By late summer of 1965, something had to be done. The problem was eventually settled in sweeping fashion and in NASA's favor: the station was transferred to NASA. Again, daily operations at the station level changed very little. All range users received support on a priority basis, allotting first priority to NASA human flight activities, second priority to Air Force ICBM launches, followed by naval fleet missile evaluation and training exercises, and lastly, support of Nuclear Test



Cultural dichotomy was very evident in this 1962 photograph of the Kano tracking station in Nigeria. Security was not an issue as the station was an object of curiosity for the local Hausa villagers. (Unnumbered photograph, Folder Number 8819, NASA Historical Reference Collection, NASA History Division, NASA Headquarters, Washington DC.)

programs. All scheduling was coordinated through the Station Director who did his best to accommodate all users. In this arrangement, all agencies eventually became very cooperative and service was virtually never denied to any legitimate user.[42]

Contractors were relied upon throughout the network. At Hawaii, the Chance Vaught Aircraft Company and its Hawaiian subsidiary, Kentron Hawaii Limited, initially played the major role. But in 1971, the operations and maintenance effort of all network stations was combined under a single contract. Bendix, already the major NASA contractor for tracking and data acquisition, was awarded the contract, one which it would keep until the station closed in 1989.[43]

Hawaii was, by all accounts, the busiest station in the worldwide network, due primarily to its location. First, launches from the Cape at a 28° inclination (that is, launches directly due east from the Cape) yielded more

visible passes at Bermuda, Carnarvon, and Hawaii than at any other station. Second, its location on a path from Vandenberg Air Force Base to the Kwajalein atoll provided an ideal location for midcourse tracking of Air Force ICBM tests. Third, PMR operations in the surface-to-air and air-to-air missile test programs required a large and rather continuous demand for radar and telemetry services. Fourth, periodic nuclear weapons testing in the South Pacific brought significant workload in support of Sandia Corporation and its subcontractors. Finally, the demands of the Pacific Fleet for support of weapons testing as an adjunct to the Vietnam War at the time called for significant resources from the station. All this yielded the heaviest workload experienced by any of the NASA stations.[44]

As the Mercury network took shape, ground station locations were determined by a number of factors. These ranged from geography to the willingness and political alliance of the local government, and to language. Perhaps no other station in the MSFN better illustrated a dichotomy in cultures than the one that sat among the shrubs and parched red clay on the outskirts of the ancient city of Kano in central Nigeria. On 19 October 1960, an agreement was reached in the capital city of Lagos, allowing the United States to construct "NASA Tracking Station 5" in the Federation of Nigeria. This news was announced with great fanfare to the Nigerians. The local Hausa villagers living around Kano described the site as "the place the Sardauna built to get the message from the stars."[45]

A NASA ground station often provided the most visible and sometimes the *only* tangible look into America's space program on foreign soil. Because of this, the station staff was often looked upon as unofficial ambassadors of international goodwill to the native populace, representing American goodwill and know-how in these countries. Kano exemplified this more than any other station. It was quite the "hi-tech" tourist attraction of its time, where open-door was the norm, drawing the curious from all parts of Nigeria. An unguarded gate leading to the station welcomed visitors. Albert E. Smith, the first Station Director said at the time, "There's nothing here that's classified; we are at home to all visitors."[46]

Life in Kano was simple, unchanged for centuries. Touareg warriors still lived in pressed-earth dwellings; naked children played along the roadside; camel caravans presided over by nomads shared the dusty washboard roads with donkey-drawn wagons and automobiles. Such was the setting for NASA's first tracking station in central Africa. Although the station would be short lived—officially closed down on 18 November 1966 just one week after Gemini 12 splashed down, a victim of technology evolution as NASA revamped the MSFN in preparation for Apollo—it met a crucial need at the time. Depending on the orbit, Kano provided anywhere from a three to six and a half minute communication window with the Mercury (and later Gemini) spacecraft as it passed over the continent after leaving the Grand Canary Island coverage area.

The station was also important because it could monitor (but not remotely trigger) a retrofire over Africa for an Indian Ocean recovery in the event of an emergency. It was operated and maintained by 30 Bendix and General Electric workers along with 10 technicians from the British Cable and Wireless Company who maintained the cross-continent and trans-Atlantic ocean floor cable between Kano and Cape Canaveral. The Air Force also had on-call a 30-person rescue team there to aid in ocean or land recovery operations if needed during Mercury and Gemini.

As the infrastructure of the MSFN was being established, international cooperation was just as crucial as it had been back in the Minitrack days. This was best illustrated at Guaymas, Mexico where diplomacy played the major role in getting the station built. The establishment of the Guaymas Station could best be described as a labored process, one that required great patience and perseverance. NASA considered a station in Mexico very important since it would be the first North American land station to establish contact, and if needed, enable the ground to command a first orbit retrofire for an Atlantic Ocean recovery in case of an emergency with the Mercury spacecraft.

In the spring of 1959, NASA presented to the State Department a list of foreign locations for tracking stations. The Department told NASA that it was reasonably optimistic about the chances to obtain entry to all areas but Mexico where, it felt, the space agency would have extreme difficulty due to internal political strife and anti-American sentiments. The Department agreed to provide assistance in every way possible in getting approval in Mexico. Initial overtures by the U.S. embassy in Mexico, however, received no response.

In the summer of 1959, Milton S. Eisenhower, advisor to his older brother President Dwight D. Eisenhower, on a trip to Mexico, made a personal appeal to President Adolfo Lopez Mateos to open negotiations. This resulted in at least an expression of interest. It was followed by discussions at a White House dinner attended by NASA Administrator Glennan during the visit of President Lopez Mateo and Mexican Ambassador Antonio Carrillo Flores later that summer. Again, interest was at least verbally expressed by the Mexican government to, at a minimum, consider the proposal. There then ensued a lengthy period of inactivity in which nothing constructive happened. NASA could not get an answer from anyone on where Mexico stood. It was emphasized by the U.S. Embassy throughout this process that the political climate in Mexico at that time was questionable at best. By then, planning of the Mercury network was well underway and the Mexican situation was beginning to impact the schedule. Great pressure came from the Langley team assigned to lead in the planning and implementation of the Mercury ground network to get an answer, either a yes or a no.

Edmond Buckley and E. J. Kerrigan made a visit to Mexico in January 1960 in a final effort to determine whether the Mexican government was really interested at all. With the very active cooperation of U.S. Embassy officials, discussions were finally held with the Acting Minister of Foreign Affairs and other top Mexican officials. The talks were deemed favorable, enough so to gain approval from President Lopez Mateo for further negotiations. Arrangements were made by Mexico City for meetings to be held between the University of Mexico and NASA on the basis of mutual scientific cooperation. This was the breakthrough that NASA had been looking for. Doing so provided the scientific impetus that Mexico deemed essential to provide the public support needed for any cooperative effort between the two countries. It was personally obvious to Edmond Buckley, though, that the actual possibility for mutually beneficial scientific cooperation of the sort desired by the Mexicans would be for projects *other than* Mercury. But he also felt that collaboration was necessary considering the importance of a Mercury station in Mexico.

Operating in this not entirely ideal but at least workable framework, negotiations proceeded over the next four months of 1960. The Mexican government presented several guidelines around which they felt the talks should be centered. These were designed primarily to assuage the prevailing negative American public sentiment:

1 Great care must be taken in all actions and public announcements so that the Mexican people not misunderstand the scientific and peaceful nature of the activity.

2 It was important that the military not participate in the operation of the station.

3 It was desirable that the activity be described as an international cooperative activity of many nations and not as a bilateral agreement between the United States and Mexico.

4 To further emphasize wide international participation, it was desirable that Australians or other nationals be present at the Mexican station; that Mexicans be present at other stations, for example, at Cape Canaveral.

5 It was preferable to have the agreement negotiated between the president of an American university and the president of the University of Mexico.[47]

A formal agreement between Mexico and the United States establishing the tracking station at Guaymas was finally signed on 12 April 1960.

With great fanfare, an inauguration ceremony officially opening the station was held 14 months later on 26 June 1961, presided by NASA Deputy Administrator Hugh Dryden, the Mexican Foreign Minister and others in a large Mexican delegation. The cost to establish the station was $2.25 million in 1960, one of the biggest projects in that area at the time.[48] The significance of an agreement establishing a spaceflight tracking station in Mexico was no small matter. Scientific and cost reasons aside, it was the first real cooperative project between the two neighboring states since before World War I and, as such, was quite momentous. Both the State Department and the U.S. Embassy made it clear that this represented a big step to bettering relations with its neighbor to the south, one that went beyond merely space exploration and Project Mercury.

One of the first actions taken during these negotiations was the establishment of the Mexico-U.S. Commission for Space Observations. The decision to establish a commission was not arrived at in a cavalier way. The U.S. State Department and General Counsel at the time required a good deal of thought before agreeing to such an arrangement, since intergovernmental commissions generally tend to establish rules which are binding on the countries involved, and thus require congressional approval. It was recognized that in this case, though, the commission would be established primarily to provide assurance to the Mexican people that this was truly a cooperative, civilian, scientific project in which their government was fully informed at the highest levels. Without the appointment of this commission, the tracking station in Mexico likely would not have materialized on schedule, if at all.

Mexico City's emphasis that, in their view, this was primarily a scientific and not political cooperative was reflected in the makeup of the commission. The Mexican Section was led by Ing. Ricardo Mongas Lopez, former Dean of the Institute of Geophysics at the University of Mexico. Mongas Lopez was a well renowned scientist with diverse experience in multiple fields who had served on many government panels. He was joined by Dr. Eugenio Mendez Docurro, Director of the National Polytechnic Institute and Ing. Jorge Suarez Dias, a former Dean of the Polytechnic Institute.

Their carefully selected counterpart, the U.S. Section, was composed of Edmond C. Buckley, Assistant Director for Space Flight Operations, NASA Headquarters; Ralph E. Cushman, Chief of Field Installations, NASA Headquarters Procurement Office; and G. Barry Graves, Head of the Langley TAGIU team responsible for planning and implementing the Mercury network. The U.S. also appointed Raymond Leddy, Counselor of the U.S. Embassy in Mexico City, as the permanent liaison officer for the U.S. Section of the commission. This last appointment was done so as to comply with Mexican wishes to have someone in their country who could handle day-to-day affairs.[49]

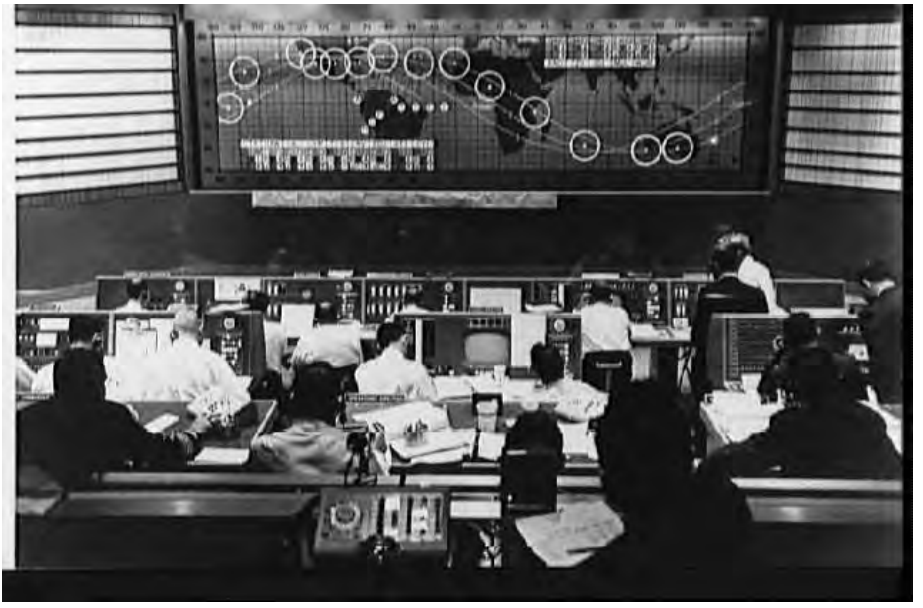
But the appointment of a commission did not translate into immediate progress. It was actually quite the contrary. Originally, meetings were to

be held every two months. In reality, however, the commission only convened when pressure from the U.S. Embassy forced its scheduling. The Mexican government still tended to proceed cautiously and kept a close watch over the actions of the commission. This caution was evident when, in the first official meeting, Ambassador Oscar Rabasa, Director in Chief for American Affairs, kept a firm control of the proceedings until he was satisfied that the direction of the meeting was headed where it was supposed to: science; only at that point did he relinquish control to Chairman Mongas Lopez. This was somewhat of a sobering process for the United States in general, and NASA in particular.[50]

Mexico has always had (and still has) a deep government bureaucracy based on Latin American heritage. Thus, NASA needed real assistance to properly work with the Mexicans so as to avoid becoming snared in bureaucratic and jurisdictional entanglement. Take for example that as many as five to seven separate Mexican bureaus ordinarily had to approve imports into the country. The use of this commission, backed by the President of Mexico and his Foreign Affairs Office, proved crucial to help cut through bureaucratic red-tape so that real work could get done. Although interbureau difficulties still arose, the United States did not become involved in them, but rather, utilized the Mexican Section of the commission to work out the often thorny issues.

In addition to dealing with governmental bureaucracy, an intangible and even more volatile issue was also at work, an issue that really called for careful diplomatic attention. Anti-American sentiment in Latin America was high at the time and strong feelings about the United States were rampant among the Mexican populace. In June 1960, Dr. Glennan invited members of the Mexican Section to Washington, at NASA's expense, to visit the Goddard, Wallops, and Langley facilities. Even though only two members came, they returned to Mexico with invaluable publicity from the traveling Mexican press. Over time, the Mexican Section of the commission turned out to be extremely helpful by giving wholehearted support to the station and keeping unfavorable press and perhaps possible demonstrations about the "United States Missile Station" at a minimum.[51]

The commission produced many favorable television and radio spots promoting the American tracking station and provided newspaper interviews in support of the project. Patience and perseverance paid off for NASA in Mexico. It was very apparent as the station finally opened that the commission (on both sides) had served its purpose well. Even Ing. Mongas Lopez and Raymond Leddy doubted that the Mexican government would ever allow it to be dissolved. "The assurance desired by both governments that the procedures be carried out in a manner that would promote good will between Mexico and the U.S. seems to have been observed," Edmond Buckley was to reflect years later on the effort it took to build up the Guaymas station. Simply put, "Different approaches were needed in different countries."[52]



The Mercury Control Center was located near the launch area at Cape Canaveral, apart from the computers which in those days were still centralized at the Goddard Space Flight Center in Maryland. Unlike the high resolution, computer graphical displays of today, the ground track of the capsule on the 'big board' was moved mechanically by wires. This picture was taken on 20 February 1962 during America's first human orbital flight, Mercury Atlas 6. (NASA Image Number 62-MA6-161)

While it was quite true that different approaches were indeed needed in different places, one common denominator generally permeated all negotiations regardless of the country involved. In setting up its network, NASA took great care to always emphasize the civilian nature of the project to the host country. The Agency's stance was that

> The National Aeronautics and Space Administration is a civilian scientific organization and that Project Mercury is a scientific experiment being conducted for the purpose of ascertaining the problems of man's existence in space. Furthermore, the results from this experiment will be made available to all the world.[53]

The promotion of the peaceful nature of the work and the sharing of results had worked well back in 1957 during the IGY when the NRL was first setting up Minitrack. Fueled by increasing East-West tensions, this

approach was needed all the more as suspicion tended to exist among many countries as to the true nature of the American space program. This was especially true of African states. Ray W. Hooker, Assistant Chief of Langley's Engineering Service Division, reported following a trip to Africa in rather blunt and sobering terms that

> In the case of the Kano and Zanzibar sites, the British have sold the local government on the fact that this is an American experiment, harmless in nature and would contribute to the scientific knowledge of the world. [But] In both the Nigerian and Zanzibar governments, there is the general native population which is capable of believing almost anything and getting quite excited about it. A rumor was circulated in Kano at the time of the site team's visit there to the effect that the team was tied in with the French atomic bomb experiment in some manner.[54]

Such gross misconceptions on the part of the local government had to be resolved diplomatically. In the Nigerian situation, personal assurance to the Emir of Kano by the Langley advance delegation was done. To do this, NASA had to enlist someone the Emir trusted, in this case Arnold W. Frutkin from the National Academy of Sciences who had established considerable positive reputation with foreign countries during the days of the IGY.[55] The American Embassy or Consulate usually served as a good starting point for such negotiations, enabling dialogue with the right officials who were in a sufficiently high position to initiate action. The case of Nigeria was especially sensitive as the British were at the time trying to end their rule in that country on good terms and most of their actions had that particular goal in mind. This turned out to benefit NASA as the British helped to clarify the peaceful intent of the Americans.

While it is entirely true that NASA's charter has always been civilian in nature, the Agency nevertheless, and from the onset, has had to work closely with the DOD to accomplish certain missions. Still true today, this was inescapable in the early 1960s: the DOD owned all the launch facilities and had a wealth of experience and knowledge in the areas of missiles and rockets. It was also in the midst of setting up its own satellite network (the Air Force Satellite Control Network, or AFSCN). Securing the "high ground" was (and is) a national security objective for the DOD.

No where else did NASA rely on the DOD more than at Cape Canaveral, Florida, where launches took place. To manage this interface, the Air Force created the position of a Department of Defense Manager for Manned Spaceflight (DDMS). DDMS was given the job of interfacing with the civilian space agency's newly created Project Mercury representatives to work through common problems. When Mercury came onboard at the Cape, the DDMS sup-

ported NASA in such areas as launch operations, range safety, contractor support services and the construction of a new control center at the launch facility.

It was in this capacity that Henry H. Clements (Associate Director of the Johnson Space Center from 1981 to 1984) began his association with NASA. As a Captain in the Air Force, he was reassigned to the DDMS, becoming the first Network Controller (NC) during the suborbital Mercury Redstone missions. "This was really a one-man operation, keeping tab on some 15 remote sites, verifying that they were ready to support the mission and that the capsule communicators stationed at various tracking sites were in position and linked to the control center at the Cape via Goddard," Clements recalled in 1982 the rudimentary setup that was Mercury Control.

> Much of the equipment and technology used in those early days was rather rudimentary. We had a display map on which the capsule was simply moved by wires and the status of our tracking stations was merely indicated by red or green lights. Our tracking stations often found it difficult to stay on the radar beacon—a beeper signal generated by the orbiting space capsule. Then, there was the problem of a smooth handover from station to station as the spacecraft circled the Earth. There were also problems when sunspots caused microwave dropouts in ground communications and pipe layers accidentally cut vital telephone cables. Yet, despite it all, we had extremely high reliability due to the outstanding support from NASA's civil service team and our contractors. They all were very anxious for this program to succeed and for our astronauts to return safely. In this effort, reliable tracking, computer and communication support was vital.[56]

The State Department was instrumental in helping to secure international cooperation not only with third-world countries but also with established United States allies. An agreement with Spain was needed, for example, for a Grand Canary Island station. A good telemetry and radar capability in the Canaries was considered critical by mission managers for orbit establishment, particularly in the case of an abort.

The Canary Islands are an archipelago of seven volcanic islands located 180 kilometers (110 miles) off the Moroccan coast of northwest Africa. These islands were known from antiquity. Prior to their conquest in 1402 by Spain, they were inhabited by the Guanches, native peoples related to the early Berbers of North Africa. The conquest of the Canaries took almost 100 years and set a notorious precedent for the conquest of the New World. Due to the terrain and staunch resistance of the native Guanches, this conquest was not completed until 1496 when the Canaries were incorporated into the Castilian kingdom. The Spanish imposed a new economic model based on single-crop

Table 3-1: The Mercury Space Flight Network (Three Orbit Configuration)[57]

Station	Abbreviation	Location	S-Band Radar	C-Band Radar	Acquisition*	Telemetry	Command	CAPCOM**
North America								
Cape Canaveral, Florida	CNV	28°28'N 80°34'W	•	•	FA	•	•	•
Corpus Christi, Texas	TEX	27°39'N 97°23'W	•		FA	•		•
Eglin AFB, Florida	EGL	30°46'N 86°53'W	•		FA			
Guaymas, Mexico	GYM	27°57'N 110°43'W	•		FA	•	•	•
Point Arguello, California	CAL	34°39'N 120°36'W	•	•	FA	•	•	•
White Sands, New Mexico	WHS	32°21'N 106°22'W		•	FA			
Australia								
Muchea, Western Australia	MUC	31°35'S 115°56'E	•		FA	•	•	•
Woomera, South Australia	WOM	31°23'S 136°53'E		•	FA	•		•
Africa								
Kano, Nigeria	KNO	12°03'N 08°31'E			SA	•		•
Zanzibar	ZZB	06°13'S 39°13'E			SA	•		•
Atlantic								
Bermuda, United Kingdom	BDA	32°15'N 64°50'W	•	•	FA	•	•	•
Grand Bahamas, British West Indies	GBI	26°38'N 78°16'W		•	M	•		•
Grand Canary, Spain	CYI	27°44'N 15°36'W	•		FA	•		•
Grand Turk, British West Indies	GTK	21°28'N 71°08'W			M	•		•
Pacific								
Canton Island, Kiribati Republic	CTN	02°50'S 171°40'W			SA	•		•
Kauai, Hawaii	HAW	22°07'N 157°40'W	•	•	FA	•	•	•
SHIPS								
Atlantic Ship (Rose Knot Victor)	ATS				FA	•		•
Indian Ocean Ship (Coastal Sentry Quebec)	CSQ				FA	•		•

* FA: fully automatic SA: semiautomatic M: manual ** CAPCOM: capsule communicator



The staff at Canary Island about 1967. Standing: John Adams, Chuck Rouillier (Station Director), Tom White, Percy Montoya, Matt Harris, Ed Bender, Ed Crough. Sitting: Clay Krugman, Dick Kelly, Roger Lee, Glenn Smith, Russ Lutz. (Photograph courtesy of Gary Schulz)

cultivation, first the sugar cane followed by wine, an all-important trade item with England. The islands eventually became an important stopping point in the trade routes with the Americas, Africa and India, and the port of La Palma turned into one of the most important ports in the Spanish Empire.

At the beginning of the twentieth century, the English introduced a new cash-crop, the banana, and the islands prospered. In 1936, Francisco Franco traveled to the Canaries as General Commandant from where he launched the military uprising of 17 July. He was able to quickly take control of the archipelago. Despite the fact that there was never actually a war in the islands during the 1940s, it was one of the places where post-war repression was most severe. Organized opposition to Franco's regime did not begin to materialize until the late 1950s, when groups such as the Spanish Communist Party and various nationalist, leftist and independence factions such as the Free Canaries Movement came onto the political scene.

When NASA approached Madrid about Grand Canary Island on 10 September 1959, Mr. William Fraleigh, First Secretary of the Consul Political Office, was told that the situation was "rather delicate." Evidently, a South American leftist newspaper that was run by exiles of the Spanish gov-

ernment had published a misleading article disclosing the fact that Madrid was in the midst of negotiating with the United States to *give* a portion of the Canary Islands in order to establish an airbase. Madrid was rather sensitive about the matter since the Spanish Moroccans were already beginning to talk about the independence of the Canaries following the example set by the French Moroccans in 1956. The Spanish government quickly advised the State Department that care must be taken to clearly delineate that the MSFN station was not going to be related in any way to the U.S. airbase already there. Otherwise, Madrid could not guarantee permission to use the station. This bit of warning was passed down through Headquarters such that by the time formal negotiations were held, the civilian versus military nature of the station project did not become the showstopper that it easily could have been.[58]

NASA always tried, to the extent possible, not to disturb the everyday lives of the local populace when building its foreign stations, but relocating the local tenants was sometimes unavoidable. This was the case on Grand Canary Island. Here, three groups of transient dwellings housing migrant farmers were located within the site of the station boundary. NASA was at first inclined to leave these dwellings alone, intending that they continue to be used after the station was up and running. But as construction progressed, it became clear that this arrangement was probably not going to be in the best interest of the Agency; that leaving the structures in their present locations and allowing people to occupy them were going to inevitably interfere with station operations. The houses had to come down.

Whenever it was necessary to displace or disrupt local property, it was NASA's standard policy to determine the removal and replacement cost prior to starting any actual work. Alternatives were pursued if the cost was deemed unacceptable. In the case of Canary Island, an agreement was reached—with the assistance of the U.S. Naval Facilities Engineering Command—to replace the structures in kind but just outside the boundary of the tracking station.[59]

★ ★ ★

Construction of the MSFN began in April 1960; by mid-summer, all stations were under construction. The last one, Kano, Nigeria, was completed in March 1961. On 1 July 1961, 24 months after awarding contracts, NASA officially accepted the new MSFN. It was quite an impressive achievement, considering that the first American human orbital flight was not scheduled for another six months. A significant amount of research and development and nonrecurring engineering went into the effort. That also included all the land and ocean-floor cable links between the GSFC core, the Cape Canaveral launch site and the individual field stations. Table 3-1

presents a summary of the MSFN as it appeared during Project Mercury, and Appendix 1 shows the network on a world map.

The standard or baseline Mercury network was set up initially to track three orbits. This was augmented during the program so that by Mercury Atlas 8 (the fifth piloted flight), missions had been extended to six orbits. In order to accommodate the additional three orbits, the Atlantic Ship was equipped with command uplink transmitters, redesignated the Pacific Command Ship (PCS) and repositioned south of Japan. Three Navy ships (*American Mariner, Huntsville and Watertown*) also supported the mission from the vicinity of Midway Island in the north-central Pacific.

These were not the only augmentations. To support the final Mercury mission—L. Gordon Cooper, Jr.'s MA-9 launched on 15 May 1962 which was 22 orbits, the standard, three-orbit Mercury network had to be modified significantly as follows:

> Indian Ocean Ship *Coastal Sentry Quebec* was moved south of Japan.
>
> Atlantic Ship *Rose Knot Victor* was moved to the South Pacific near Easter Island.
>
> DOD ship *Range Tracker* was stationed northeast of Midway Island.
>
> DOD ship *Twin Falls Victory* was stationed between Bermuda and the U.S. East Coast.
>
> The DOD Eastern Test Range station on Ascension Island and Puerto Rico provided FPS-16 radar and telemetry recording.
>
> The new MSFN Gemini/Apollo station at Antigua was turned on to record telemetry and voice.
>
> Temporary voice communication sites were erected on Kwajalein Island, San Nicholas Island and Wake Island.

DOD aircraft were assigned to provide voice and telemetry relays. Also starting with this mission, the radio link between Bermuda and Cape Canaveral was supplanted by submarine cable and the GSFC computers were upgraded to IBM 709s.[60]

Stations had different roles and different equipment depending on their roles. Not all stations used flight controllers for instance. Sites like White Sands and Eglin were used only for tracking. The primary stations with full TT&C capabilities were the Mercury Control Center (MCC) at Cape Canaveral, Florida; Bermuda; Muchea, Western Australia; Kauai, Hawaii; Point Arguello,

California; and Guaymas, Mexico. At these stations, commands—still few in number—could be transmitted up to the spacecraft as combination of tone pulses. Telemetry was received, demodulated and displayed in the form of strip charts and analog meter readings for flight controller evaluation. They read the health and status of the spacecraft, for example, marking readouts with different color grease pencils, each denoting the different times at which readings were taken. Between passes, summaries and reports were sent to the MCC via teletype; few telephone voice circuits were available at the time.[61]

The relative importance of a given site varied depending on what was happening during a flight. Cape Canaveral and Grand Bahama, for example, provided vital telemetry, tracking, command, and voice coverage during launch and ascent. Conversely, Grand Turk Island provided radar tracking during Atlantic Ocean reentries. Bermuda bore the responsibility of deter-



Panoramic view of Launch Complex 14 as Mercury Atlas 9 is readied for flight in this picture taken eight days before its launch as the last Mercury mission. (NASA Image Number GPN-2000-000609)

mining whether a mission should continue or be aborted. Should Bermuda order a "no-go", Grand Canary Island then provided the crucial tracking needed for an emergency reentry. Finally, Kauai could transmit command tones up to the spacecraft to set off the timers used to fire deorbit retrorockets in the event of crew incapacitation.[62]

One capability that was common to just about all the stations was, as one would expect, air-to-ground voice communications. Not all stations used astronaut Capcoms, though (which by tradition is always an astronaut).[63] For stations where a NASA Capcom could not be assigned, a Communication Technician (or Comm Tech) handled voice calls to the astronaut. Before September 1962, the "Original 7" were all the astronauts that NASA had, so being able to speak with an astronaut on a space mission in those pioneering days was beyond privilege and quite an honor. The communications console at the Muchea station, for instance, was memorialized for years with a plaque which read:

> This plaque is to mark the spot where an Australian first spoke to a space traveler. The Australian was the communication technician at Muchea, Mr. Gerry O'Connor, and the space traveler was astronaut John Glenn.[64]

A key to maintaining network reliability was the utilization of built-in operational redundancy using spares and backup equipment (plus frequent exercise of that equipment). During the first piloted orbital flight on Mercury Atlas 6, controllers worst fears came true when the prime computer used for orbit determination at GSFC "crashed." But the built-in redundancy of the system had another computer available which was immediately brought online. Station personnel and equipment were kept sharp through exercises, simulations and frequent system checkouts. Although not part of the active network of stations when it first started, Wallops Island, just off the Virginia coast, served as a personnel training and network equipment test facility. This network training center was established by GSFC in July 1961 shortly after its networks became operational. There, equipment was prepared, tested and checked-out prior to shipment and field installation. At Wallops, all NASA and most of the contractor supervisory, maintenance and operation personnel were trained before their deployment overseas. Students at the Center were, for the most part, new engineering college graduates or electronics technicians in their 20s or early 30s. By 1963, the final year of Project Mercury, enrollment at the Center had reached 255 students.[65]

★ ★ ★

A hallmark of the NASA spaceflight tracking networks has been the simple but distinguishing fact that a mission has never been compromised

because of network problems, an enduring value that Covington espoused to the men and women who worked under him. The MSFN, nevertheless, did have its share of drama during Mercury:

> Mercury Atlas 5 (MA-5): On the flight of Enos the chimpanzee, a tractor accidentally plowed up cable outside Tucson, Arizona, cutting off communications with the Hawaii and California stations. Unfortunately, the alternate DOD circuit that had been provided in case something like this happened was also severed at the same time. To compound the situation, telemetry revealed that fuel was literally boiling away from the capsule, thus vapors—not fuel—was getting to the spacecraft thrusters needed to control the orientation of the spacecraft in the weightlessness of space. An inordinate amount of fuel was being consumed based on the data obtained by the MSFN. Mission Control decided to cut the flight short by one orbit and commanded retrorocket fire from Guaymas (exactly what the station was there for). AT&T, who owned the ground cables, worked frantically to come up with a solution. Just as the spacecraft approached North America, NASA communications traffic was rerouted around the cable break. Mission Control had just 12 seconds left in the window to initiate reentry. Enos was saved.
>
> MA-6: As mentioned before, during astronaut John H. Glenn, Jr.'s pioneering flight, the backup computer at GSFC came in handy, performing computations for three minutes while the prime computer was being rebooted.
>
> MA-7: Using radar data from Point Arguello, California immediately following retrofire, the Goddard computers correctly determined that astronaut M. Scott Carpenter had overshot the intended splashdown point by 386 kilometers (240 miles). Mercury Control was at first skeptical when informed of this and the news was received with disbelief. Subsequent network tracks at Whites Sand and Corpus Christi confirmed Goddard's initial prediction of the overshoot.
>
> MA-8: Solar activity was high and serious communication problems were experienced by the MSFN. Navy instrumentation ships were also used to augment the ground stations. On the next to last orbit, a power failure occurred at Point Arguello and a backup DOD site on San Nicholas Island was used to provide radar tracking. As an experiment, astronaut Walter M. Schirra, Jr. turned off the tracking beacon on his spacecraft during the fourth and part of the fifth orbit to see if ground radars could skin-track the Mercury capsule reliably. They could not.

> A problem also developed on the flight with the onboard thermal control system that was due to, of all things, wax buildup. The problem was serious enough that after the first orbit, MCC had to decide whether to bring Schirra back or give the problem a chance to correct itself on the next orbit, which it did. During this critical period, flight controllers had a constant flow of data on which mission-critical decisions were made for example, temperature measurements from the spacecraft to assure them that Schirra would not become dehydrated by excessive heat.
>
> MA-9: On the last flight of Project Mercury, 15 incidents of radio interference were reported by the network. Bermuda, for example, reported hearing the Voice of America and that of a Greenville, North Carolina, amateur radio operator. DOD frequency controllers were able to contact the offending operator and clear the NASA channels. On several occasions, however, amateur radio operators actually succeeded in contacting astronaut Gordon Cooper! (He was directed by Mercury Control Center not to respond.) More serious were instances where industrial equipment generated excessive radio interference around ground stations. NASA, for the most part, was able to convince local equipment owners to shut off the equipment on station passes during the 34 hour flight.[66]

Mercury Atlas 9 splashed down on 16 May 1963, bringing America's first human space adventure to a successful end. Particular praise for the unsung heroes of the invisible network came from Cooper himself at the Project Mercury summary conference in October 1963. When asked about his experience and the people who made the program a success, Cooper paused and then replied:

> It would be difficult to single out any one group or organization for special praise because they were all a wonderful team and a smooth blending of extraordinarily competent technical skills. However, I do think that if one could be mentioned organizationally, certainly I would have mentioned the worldwide network. It is certainly comforting to know when you are out there, that the world's finest communications network and the finest electronic facilities that man can devise are functioning with a fantastic computer complex that will allow the onboard systems specialists to break out their diagrams and tell you immediately what your situation is in the event of trouble, and this is indeed what happened on several occasions. Without this marvelous organization, it might have been a little more difficult to get back home.[67]

By the end of Project Mercury, there was little doubt that the MSFN had more than met the expectations of its designers. In a span of less than five years, NASA's manned spaceflight network went from a concept on the drawing board of the Tracking System Study Group to a $125 million global network that safely brought four Americans back from orbit. This price tag—of which $53 million went into network facilities and $72 million to operating cost—came out to just under one-third of the $400 million total that NASA spent on Project Mercury. The yearly budget just to operate NASA's tracking stations stood at $225 million. To illustrate the changing times, just six years earlier, the entire Minitrack Network was constructed and operated for a mere $13 million.[68]

As 1963 turned into 1964, Goddard's two networks, the STADAN and the Manned Space Flight Network (note the name change from "Mercury" to "Manned") were a key and proven part of the five year old space agency. Workforce at the GSFC now accounted for 11 percent of the NASA total. Stations had been established spanning Alaska to Australia. Conceived as a "manned-satellite" program, Project Mercury was just the beginning. It allowed America to not only send astronauts into space for the first time, but also to man-rate its worldwide tracking and data acquisition network. This network matured quickly to become the dependable safeguard that NASA was counting on. Changes would soon be coming as the United States picked up its pace, sending more astronauts into Earth orbit and beyond in a determined race that would culminate with humans on the Moon.